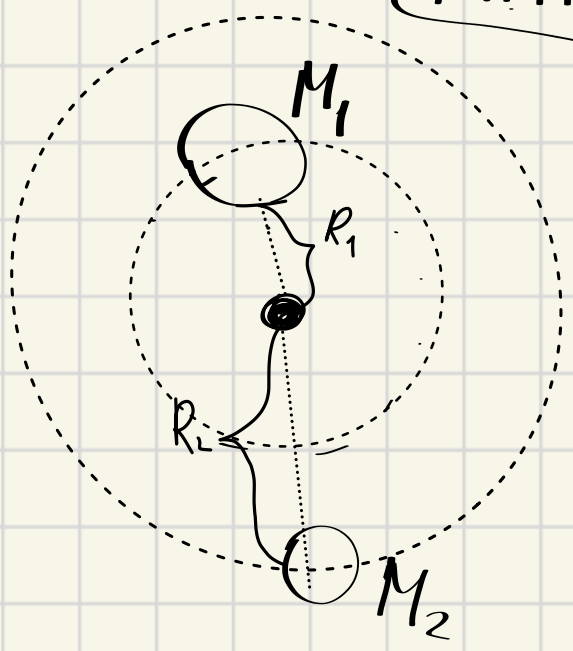
Силы упругости. Силы трения

$M_1 + M_2 = M$



① $F_{тяг1} = G\frac{M_1 M_2}{L} = G\frac{M_1 M_2}{(R_1+R_2)^2}$

$F_{тяг1} = m_1 a_y = M_1 \omega^2 R_1$

$\Big\} \frac{GM_2}{(R_1+R_2)^2} = \omega^2 R_1$

② $F_{тяг2} = \frac{G M_1 M_2}{(R_1+R_2)^2}$

$F_{тяг2} = m_2 a_y = M_2 \omega^2 R_2$

$\Big\} \frac{GM_1}{(R_1+R_2)^2} = \omega^2 R_2$

Сложим: $\frac{G(M_1+M_2)}{(R_1+R_2)^2} = \omega^2 (R_1+R_2)$

M

$\omega = \frac{2\pi}{T}$

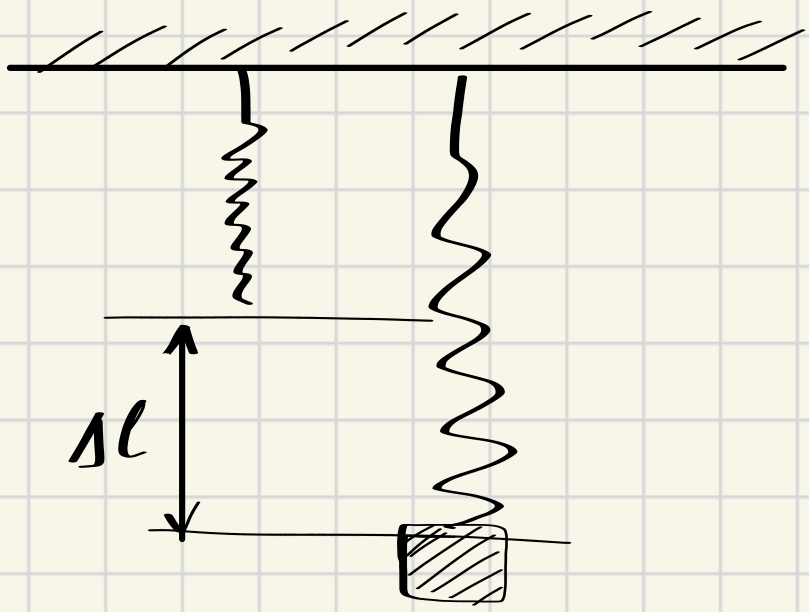
$GM = \frac{4\pi^2}{T^2}(R_1+R_2)^3$

$$T^2 = \frac{4\pi^2 (R_1+R_2)^3}{GM}$$

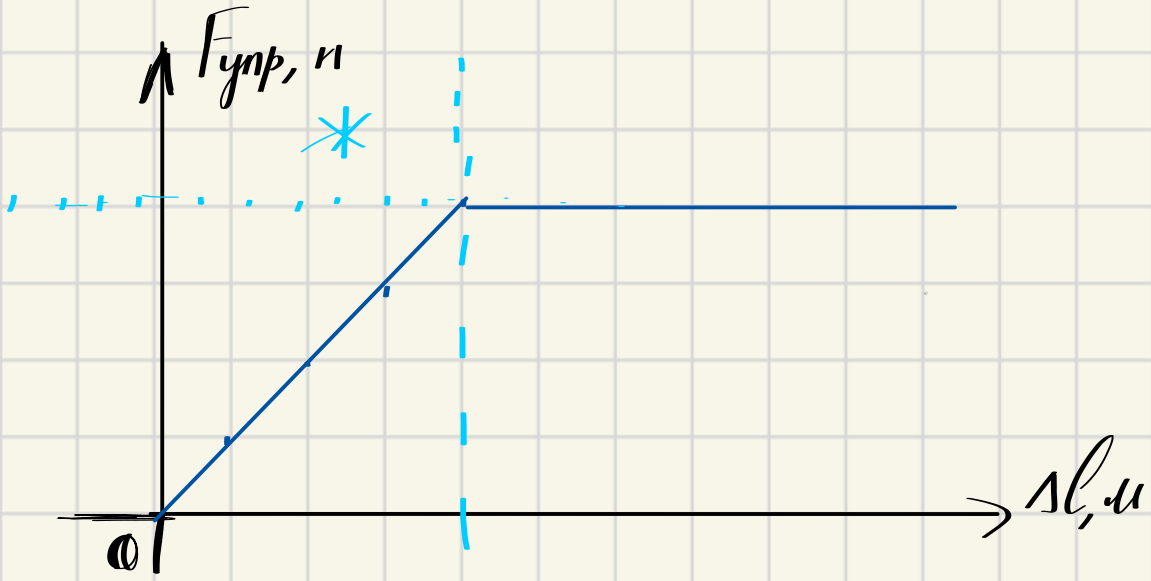
Сила упругости

$F_{упр} \sim \Delta\ell$

$$F_{упр} = k\Delta\ell$$

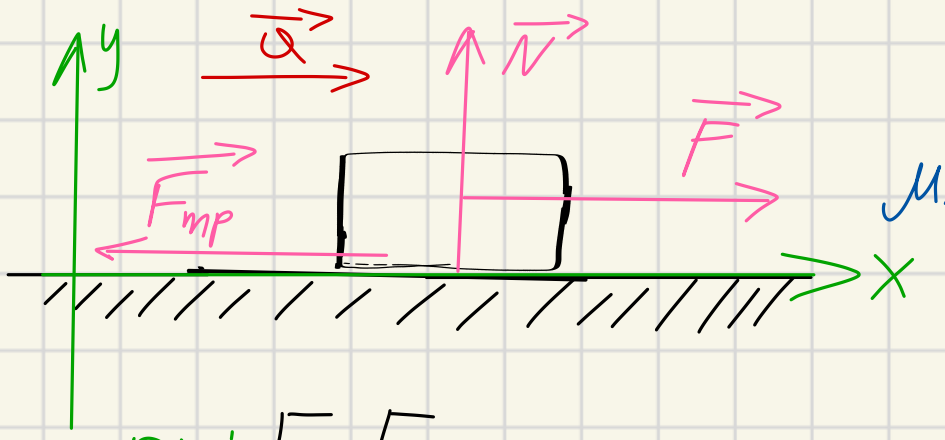


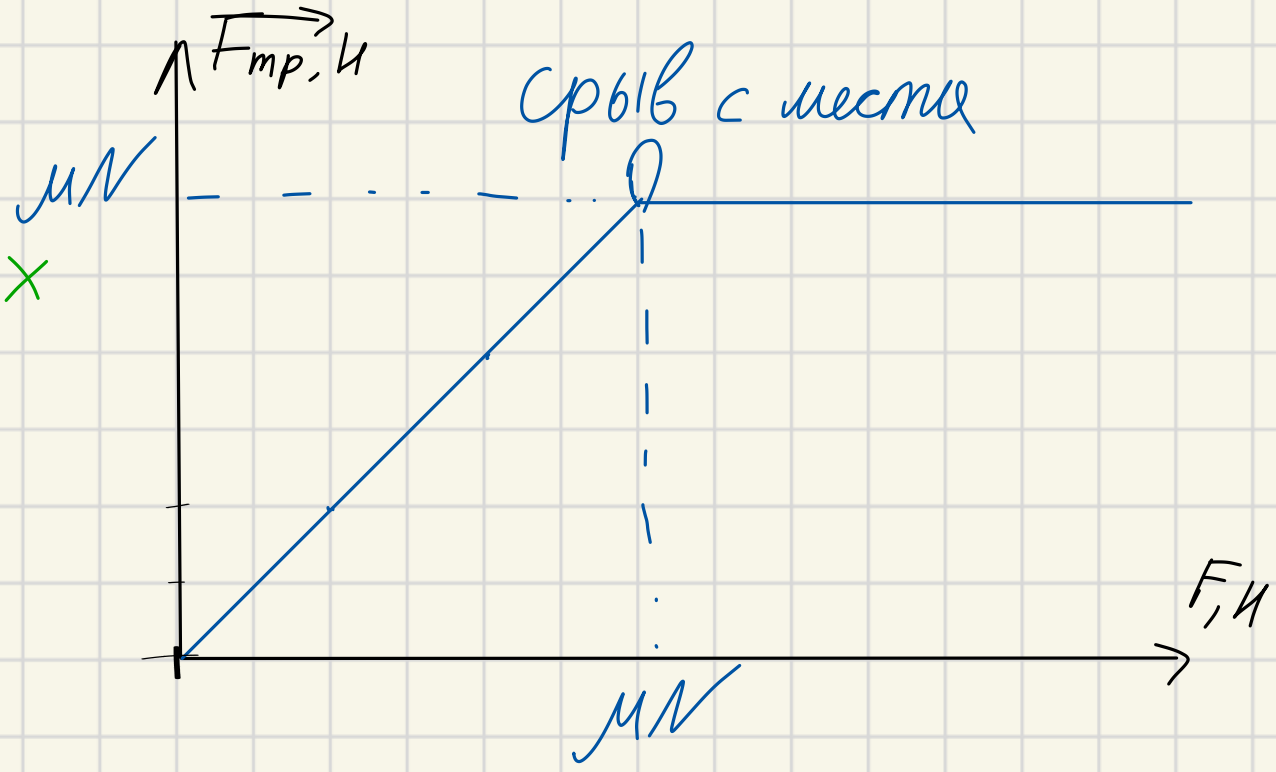
k - жесткость $\quad [k] = \frac{Н}{м}$



✳ упругая деформация

Силы трения





OX: $F - F_{тр} = ma$

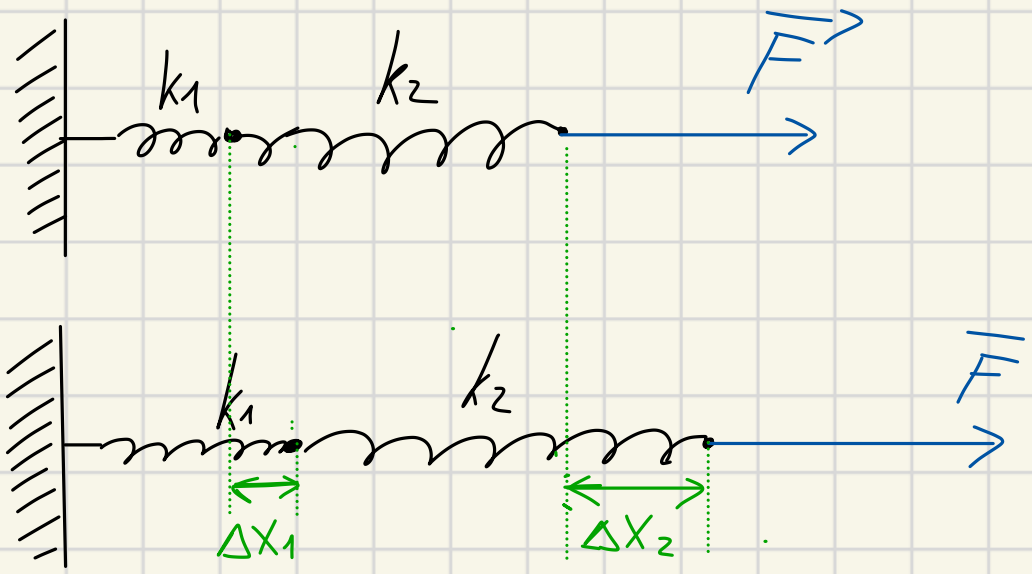
OY: $N - mg = 0$

$N = mg$

$F - \mu mg = ma$

$k_{экв} - ?$

Задача 1



$\Delta X = \Delta X_1 + \Delta X_2$

$F_1 = F_2 = F$

$$\begin{cases} F = k_{экв} \cdot \Delta X \\ F_1 = k_1 \cdot \Delta X_1 \\ F_2 = k_2 \Delta X_2 \end{cases} \rightarrow \begin{cases} \Delta X = \dfrac{F}{k_{экв}} \\ \Delta X_1 = \dfrac{F_1}{k_1} \\ \Delta X_2 = \dfrac{F_2}{k_2} \end{cases}$$

$$\frac{F}{k_{экв}} = \frac{F_1}{k_1} + \frac{F_2}{k_2} = \frac{F}{k_1} + \frac{F}{k_2}$$

$$\frac{1}{k_{экв}} = \frac{1}{k_1} + \frac{1}{k_2}$$ – Ответ

Задача 2

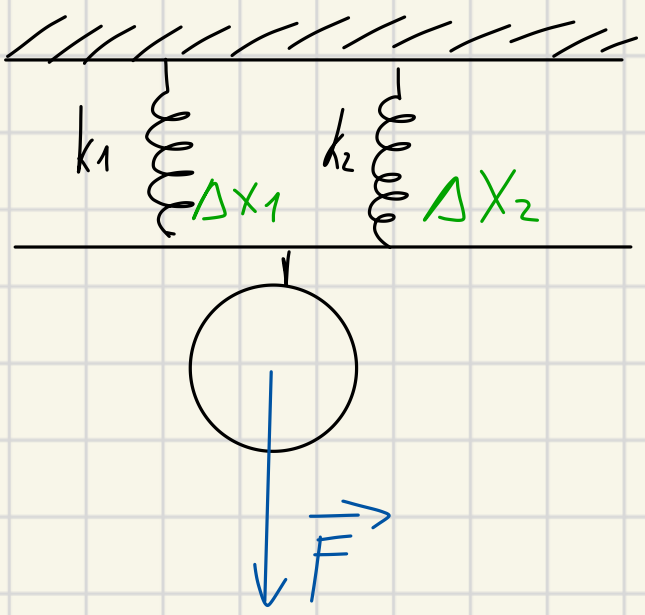
$\Delta X_1 = \Delta X_2 = \Delta X$

$F = F_1 + F_2$

$$\begin{cases} F = \Delta X k_{экв} \\ F_1 = \Delta X_1 \cdot k_1 \\ F_2 = \Delta X_2 \cdot k_2 \end{cases}$$

$k_{экв} \cdot \Delta X = k_1 \Delta X + k_2 \cdot \Delta X$

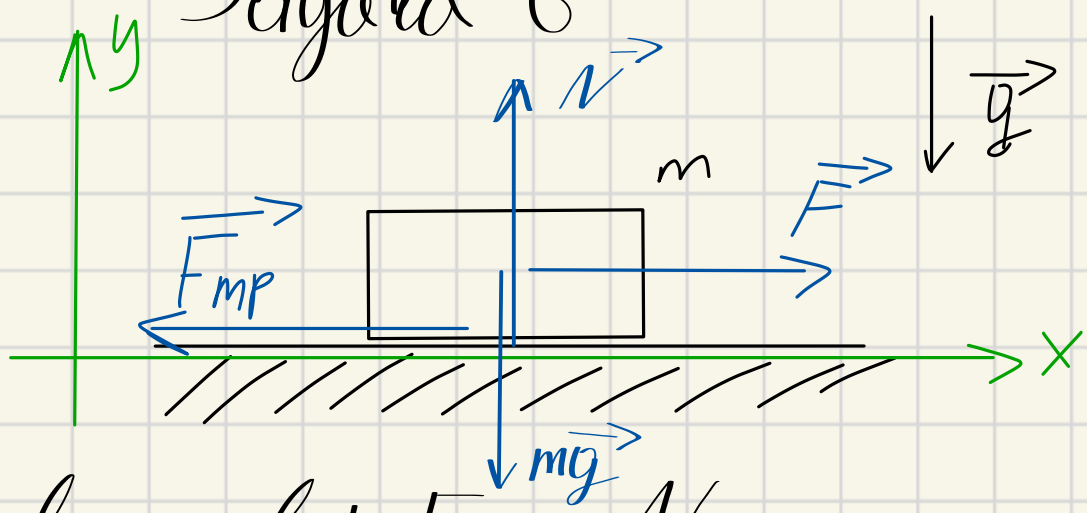
$k_{экв} = k_1 + k_2$



Задача 6

$m = 5$ кг
$\mu = 0{,}2$
$g = 10 \frac{Н}{кг}$
а) $F_1 = 9$ Н
б) $F_2 = 15$ Н

$F_{тр} - ?$



Условие срыва: $F = \mu N$

$(F_{тр})_{max} = \mu N = \mu mg$

По II ЗН: oy: $N = mg$

$(F_{тр})_{max} = 0{,}2 \cdot 5 \cdot 10 = 10$ Н

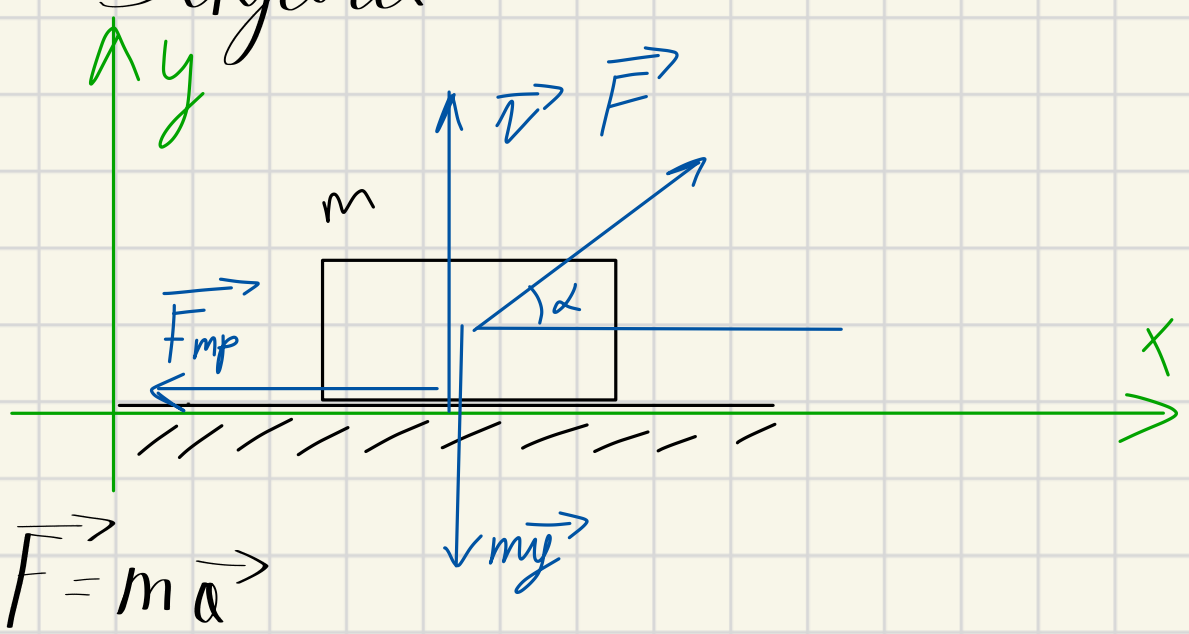
а) $(F_{тр})_{max} > F_1 \Rightarrow F_{тр} = F_1 = 9$ Н

б) $F > (F_{тр})_{max} \Rightarrow F_{тр} = (F_{тр})_{max} = 10$ Н

Задача 7

$m = 10$ кг
$\alpha = 30^\circ$
$F = 60$ Н
$g = 10 \frac{м}{с^2}$

$\mu - ?$



II ЗН $\sum \vec{F} = m\vec{a}$

ox: $F_{тр} = F\cos\alpha$; $\mu N = F\cos\alpha$

oy: $N = mg - F\sin\alpha$

$\mu(mg - F\sin\alpha) = F\cos\alpha$